DATASHEET

# HALCON Calibration Plate, Aluminum Composite

HALCON calibration plate with rectangularly arranged marks, the layout that HALCON's `gen_caltab` operator describes: black circular marks of one size in a regular grid on a white field, inside a black frame with a triangular mark in one corner. The frame lets `find_caltab` locate the plate and the triangle fixes its orientation. For use with `find_calib_object`, `find_caltab` and `calibrate_cameras`.

## 1 Ordering

| Part number | Size (W × H) |
|---|---|
| CT-HAL-P-200X150-STD | 200 × 150 mm |
| CT-HAL-P-300X200-STD | 300 × 200 mm |
| CT-HAL-P-400X300-STD | 400 × 300 mm |
| CT-HAL-P-600X400-STD | 600 × 400 mm |
| CT-HAL-P-800X600-STD | 800 × 600 mm |

## 2 Key specifications

| ACCURACY | SUBSTRATE | FLATNESS |
|---|---|---|
| ±0.3 mm + 0.2 mm/m (any mark center to nominal grid, at 20 °C) | Aluminum / LDPE / aluminum composite, 6 mm | ≤ 0.5 mm per meter, board supported on three points |

Unless stated otherwise, dimensions are nominal at 20 °C.

## 1 Specifications · geometry

| FEATURE | SPECIFICATION |
|---|---|
| Feature position tolerance | ±0.3 mm + 0.2 mm/m (any mark center to nominal grid, at 20 °C) |
| Mark pitch, mean | ±0.05 mm |
| Circularity | ≤ 0.05 mm radial deviation |
| Edge radius | R6 mm |

## 2 Specifications · physical

| FEATURE | SPECIFICATION |
|---|---|
| Substrate | Aluminum / LDPE / aluminum composite, 6 mm |
| Weight | 6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg) |
| Pattern | Direct UV print, matte, black on white; optical density ≥ 1.6 |
| Flatness | ≤ 0.5 mm per meter, board supported on three points |
| Mounting | 4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size |

## 3 Specifications · environment

| FEATURE | SPECIFICATION |
|---|---|
| Operating temperature | –50 to +80 °C |
| Thermal expansion | ≈ 24 µm/(m·°C) |
| Storage | –20 to +60 °C, dry |
| Outdoor lifetime | up to 3 years (UV-stable print) |
| Cleaning | water or ≤ 70 % isopropyl alcohol, soft cloth |

## 4 Deliverables

| INCLUDED | | SERVICE | |
|---|---|---|---|
| Measurement | Every board is measured before it ships (dimensional, flatness, visual) | Lead time | Made to order · ships in 2–3 weeks |
| Files | `.halcon.descr`, `.opencv.json`, `.dxf` | Options | Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE |
| Documents | this datasheet, CT-UM-GEN user manual | | |

## 5 Compliance & customs

| TOPIC | STATEMENT |
|---|---|
| RoHS | RoHS-compliant materials. |
| HS code | `9031 80` Not a toy. not for food contact. |

## 6 Revision history

| REV | DATE | CHANGE |
|---|---|---|
| A | 2026-10 | First issue, planning values |

## 7 Disclaimer

Values are nominal unless stated as measured.

Product configuration, lead time and regulatory statements remain subject to the confirmations identified in this datasheet.

calib.tools/files/CT-DS-HAL-P